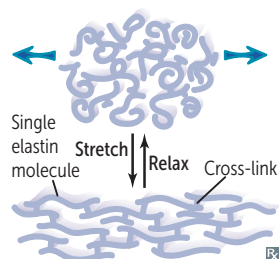


Elastin

Stretchy protein within skin, lungs, large arteries, elastic ligaments, vocal cords, epiglottis, ligamenta flava (connect vertebrae → relaxed and stretched conformations). Rich in nonhydroxylated proline, glycine, and lysine residues, vs the hydroxylated residues of collagen.

Tropoelastin with fibrillin scaffolding.

Cross-linking occurs extracellularly via lysyl oxidase and gives elastin its elastic properties.

Broken down by elastase, which is normally inhibited by α_1 -antitrypsin.

α_1 -Antitrypsin deficiency results in unopposed elastase activity, which can cause COPD.

Marfan syndrome—autosomal dominant (with variable expression and symptoms due to pleiotropy) connective tissue disorder affecting skeleton, heart, and eyes. *FBN1* gene mutation on chromosome 15 (fifteen) results in defective fibrillin-1, a glycoprotein that forms a sheath around elastin and sequesters TGF- β . Findings: tall with long extremities; chest wall deformity (pectus carinatum [pigeon chest] or pectus excavatum **A**); hypermobile joints; long, tapering fingers and toes (arachnodactyly); cystic medial necrosis of aorta; aortic root aneurysm rupture or dissection (most common cause of death); mitral valve prolapse; ↑ risk of spontaneous pneumothorax.

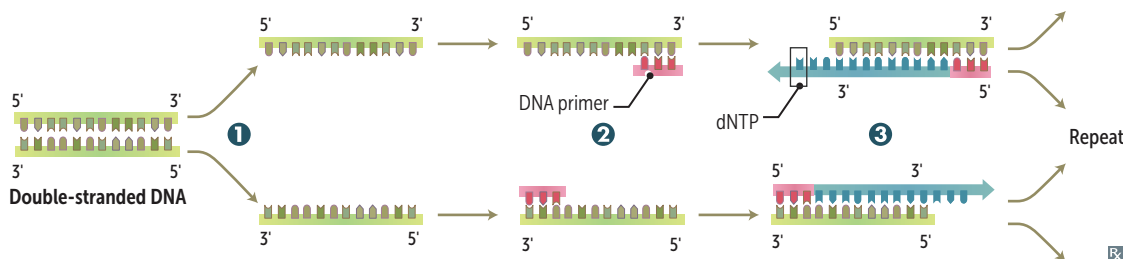
Homocystinuria—most commonly due to cystathionine synthase deficiency leading to homocysteine buildup. Presentation similar to Marfan syndrome with pectus deformity, tall stature, ↑ arm:height ratio, ↓ upper:lower body segment ratio, arachnodactyly, joint hyperlaxity, skin hyperelasticity, scoliosis, fair complexion (vs Marfan syndrome).

	Marfan syndrome	Homocystinuria
INHERITANCE	Autosomal dominant	Autosomal recessive
INTELLECT	Normal	Decreased
VASCULAR COMPLICATIONS	Aortic root dilatation	Thrombosis
LENS DISLOCATION	Upward/temporal (Marfan fans out)	Downward/nasal

► BIOCHEMISTRY—LABORATORY TECHNIQUES

Polymerase chain reaction

Molecular biology lab procedure used to amplify a desired fragment of DNA. Useful as a diagnostic tool (eg, neonatal HIV, herpes encephalitis).



1 Denaturation—DNA template, DNA primers, a heat-stable DNA polymerase, and deoxynucleotide triphosphates (dNTPs) are heated to separate the DNA strands.

2 Annealing—sample is cooled. DNA primers anneal to the specific sequence to be amplified on the DNA template.

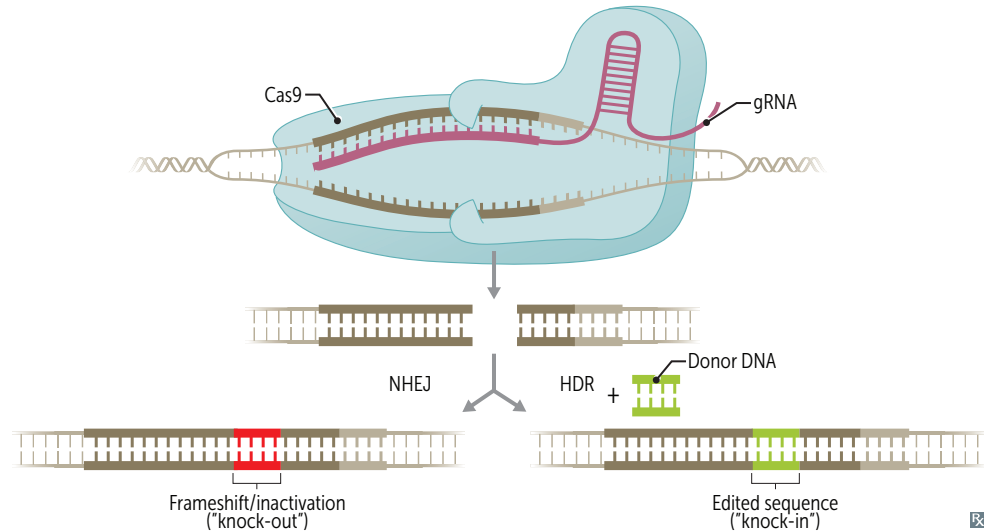
3 Elongation—temperature is increased. DNA polymerase adds dNTPs to the strand to replicate the sequence after each primer.

Heating and cooling cycles continue until the amount of DNA is sufficient.

CRISPR/Cas9

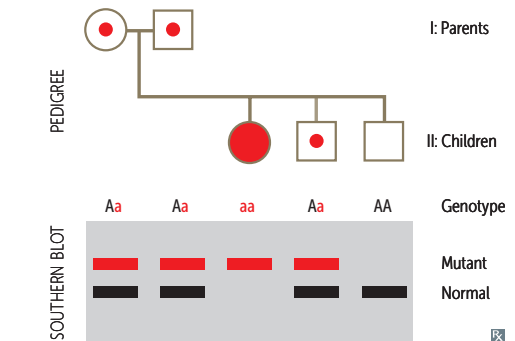
A genome editing tool derived from bacteria. Consists of a guide RNA (gRNA), which is complementary to a target DNA sequence, and an endonuclease (Cas9), which makes a single- or double-strand break at the target site.

Applications include removing virulence factors from pathogens, replacing disease-causing alleles of genes with healthy variants (eg, sickle cell disease), and specifically targeting tumor cells.

**Blotting procedures****Southern blot**

1. DNA sample is enzymatically cleaved into smaller pieces, which are separated by gel electrophoresis, and then transferred to a membrane.
2. Membrane is exposed to labeled DNA probe that anneals to its complementary strand.
3. Resulting double-stranded, labeled piece of DNA is visualized when membrane is exposed to film or digital imager.

Useful to identify size of specific sequences (eg, determination of heterozygosity [as seen in image], # of CCG repeats in *FMRI* to diagnose fragile X syndrome).

**SNoW DRoP:**

Southern = **D**N

Northern = **R**NA

Western = **P**rotein

Northern blot

Similar to Southern blot, except that an RNA sample is electrophoresed. Useful for studying mRNA levels and size, which are reflective of gene expression. Detects splicing errors.

Western blot

Sample protein is separated via gel electrophoresis and transferred to a membrane. Labeled antibody is used to bind relevant protein. This helps identify specific protein and determines quantity.

Flow cytometry

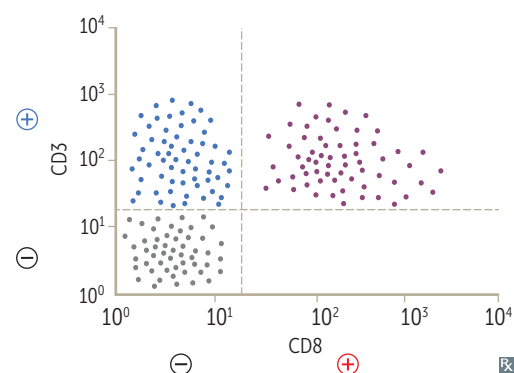
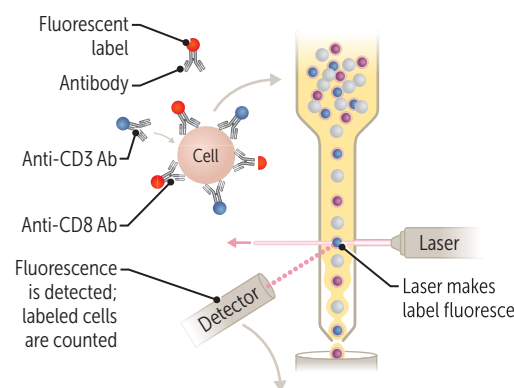
Laboratory technique to assess size, granularity, and protein expression (immunophenotype) of individual cells in a sample.

Cells are tagged with antibodies specific to surface or intracellular proteins. Antibodies are then tagged with a unique fluorescent dye. Sample is analyzed one cell at a time by focusing a laser on the cell and measuring light scatter and intensity of fluorescence.

Data are plotted either as histogram (one measure) or scatter plot (any two measures, as shown). In illustration:

- Cells in left lower quadrant $\ominus$ for both CD8 and CD3.
- Cells in right lower quadrant $\oplus$ for CD8 and $\ominus$ for CD3. In this example, right lower quadrant is empty because all CD8-expressing cells also express CD3.
- Cells in left upper quadrant $\oplus$ for CD3 and $\ominus$ for CD8.
- Cells in right upper quadrant $\oplus$ for both CD8 and CD3.

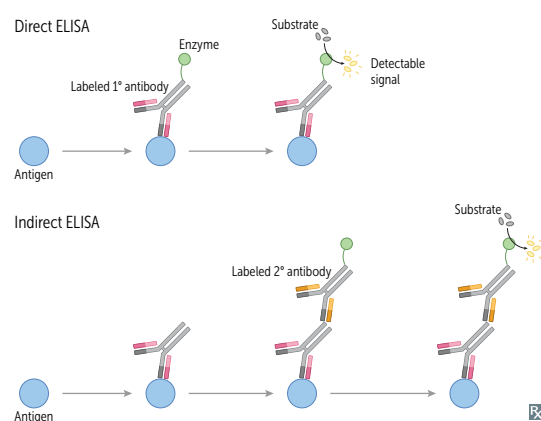
Commonly used in workup of hematologic abnormalities (eg, leukemia, paroxysmal nocturnal hemoglobinuria, fetal RBCs in pregnant person's blood) and immunodeficiencies (eg, CD4+ cell count in HIV).

**Microarrays**

Used to compare the relative transcription of genes in two RNA samples. Can detect single nucleotide polymorphisms (SNPs) and copy number variants (CNVs) for genotyping, clinical genetic testing, forensic analysis, and cancer mutation and genetic linkage analysis when DNA is used.

Enzyme-linked immunosorbent assay

Immunologic test used to detect the presence of either a specific antigen (in direct ELISA) or antibody (in indirect ELISA) in a patient's blood sample. Detection involves the use of an antibody linked to an enzyme. Added substrate reacts with the enzyme, producing a detectable signal. Can have high sensitivity and specificity, but is less specific than Western blot. Often used to screen for HIV infection.



Karyotyping

Colchicine is added to cultured cells to disrupt the assembly of mitotic spindles and arrest cells at mitosis. Chromosomes are stained, ordered, and numbered according to morphology, size, arm-length ratio, and banding pattern (arrows in **A** point to extensive abnormalities in a cancer cell).

Can be performed on a sample of blood, bone marrow, amniotic fluid, or placental tissue.

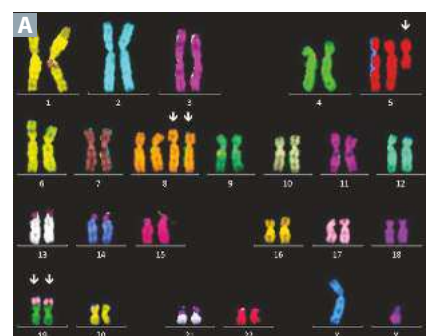
Used to diagnose chromosomal imbalances (eg, autosomal trisomies, sex chromosome disorders).

**Fluorescence in situ hybridization**

Fluorescent DNA or RNA probe binds to specific gene or other site of interest on chromosomes.

Used for specific localization of genes and direct visualization of chromosomal anomalies.

- Microdeletion—no fluorescence on a chromosome compared to fluorescence at the same locus on the second copy of that chromosome.
- Translocation—**A** fluorescence signal that corresponds to tetrasomy (chromosome 8), gain of chromosome (chromosome 5), and unbalanced translocation (between chromosomes 17 and 19).
- Duplication—a second copy of a chromosome, resulting in a trisomy or tetrasomy (eg, chromosome 8 in **A**).

**Molecular cloning**

Production of a recombinant DNA molecule in a bacterial or eukaryotic cell line host. Useful for production of human proteins in bacteria (eg, human growth hormone, insulin).

Gene expression modifications

Transgenic strategies in mice involve:

- Random insertion of gene into mouse genome
- Targeted insertion or deletion of gene through homologous recombination with mouse gene

Knock-**out** = removing a gene, taking it **out**.

Knock-**in** = **in**serting a gene.

Random insertion—constitutive expression.

Targeted insertion—conditional expression.

RNA interference

Process whereby small non-coding RNA molecules target mRNAs to inhibit gene expression.

MicroRNA

Naturally produced by cell as hairpin structures. Loose nucleotide pairing allows broad targeting of related mRNAs. When miRNA binds to mRNA, it blocks translation of mRNA and sometimes facilitates its degradation.

Abnormal expression of miRNAs contributes to certain malignancies (eg, by silencing an mRNA from a tumor suppressor gene).

Small interfering RNA

Usually derived from exogenous dsRNA source (eg, virus). Once inside a cell, siRNA requires complete nucleotide pairing, leading to highly specific mRNA targeting. Results in mRNA cleavage prior to translation.

Can be produced by transcription or chemically synthesized for gene “knockdown” experiments.

► BIOCHEMISTRY—GENETICS

Genetic terms

TERM	DEFINITION	EXAMPLE
Codominance	Both alleles contribute to the phenotype of the heterozygote.	Blood groups A, B, AB; α_1 -antitrypsin deficiency; HLA groups.
Variable expressivity	Patients with the same genotype have varying phenotypes.	Two patients with neurofibromatosis type 1 (NF1) may have varying disease severity.
Incomplete penetrance	Not all individuals with a pathogenic gene variant show the disease. % penetrance $\times$ probability of inheriting genotype = risk of expressing phenotype.	BRCA1 gene mutations do not always result in breast or ovarian cancer.
Pleiotropy	One gene contributes to multiple phenotypic effects.	Cystic fibrosis manifests with thick mucus in the lungs and GI tract, pancreatic insufficiency, male infertility.
Anticipation	Increased severity or earlier onset of disease in succeeding generations.	Trinucleotide repeat diseases (eg, Huntington disease).
Loss of heterozygosity	If a patient inherits or develops a mutation in a tumor suppressor gene, the wild type allele must be deleted/mutated/eliminated before cancer develops. This is not true of oncogenes.	Retinoblastoma and the “two-hit hypothesis,” Lynch syndrome (HNPCC), Li-Fraumeni syndrome.
Epistasis	The allele of one gene affects the phenotypic expression of alleles in another gene.	Albinism, alopecia.
Aneuploidy	An abnormal number of chromosomes; due to chromosomal nondisjunction during mitosis or meiosis.	Down syndrome (trisomy 21), Turner syndrome (45,XO), oncogenesis.